

ELECTRICITY – CIRCUITS, CURRENT & RESISTANCE

KEY VOCABULARY

Current (I): the rate of flow of charge, measured in amperes (A).

Potential difference (V): energy transferred per unit charge, in volts (V).

Resistance (R): how much a component opposes the current, in ohms (Ω).

Charge (Q): measured in coulombs (C); $Q = I \times t$.

COMMON MISCONCEPTIONS

~~Current is used up by components.~~

✓ Current is the same all around a series circuit.

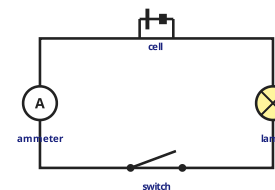
~~An ammeter is connected in parallel.~~

✓ An ammeter is connected in series.

~~More resistance means more current.~~

✓ More resistance means less current (for fixed V).

A SIMPLE SERIES CIRCUIT



Ammeter in **series**; voltmeter would go in **parallel**.

1 OHM'S LAW

A resistor has a p.d. of 6 V and a current of 0.5 A. **Calculate** its resistance. [2 marks]

Answer: _____ Ω

2 CURRENT IN SERIES

State what happens to the **current** at each point in a series circuit. [1 mark]

3 CHARGE FLOW

A current of 3 A flows for 120 s. **Calculate** the charge that flows ($Q = I \times t$). [2 marks]

Answer: _____ C

4 SERIES VS PARALLEL

Explain one advantage of connecting lamps in parallel rather than series. [2 marks]

5 MEASURING

Complete the sentence using the word bank. [2 marks]

An _____ is placed in series; a voltmeter is placed in _____.

series

parallel

ammeter

voltmeter

6 RESISTOR I-V GRAPH

Describe the I-V graph for a fixed resistor at constant temperature. [2 marks]

